

The Graph Theory Behind Games: The Seven Bridges of Königsberg and the Game of Hex

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1 The Problem of the Seven Bridges of Königsberg

1.1 Introduction

During the 18th century, the city of Königsberg (currently Kaliningrad, Russia) was divided into four districts by the Pregel River. Seven bridges connected the four districts (as shown below). Legend had it that residents would stroll throughout the city and eventually challenged each other to find a way to traverse all seven bridges only one time each.

In other words, the problem was to design a walking tour of Königsberg (through all its four districts) by crossing all seven bridges exactly once.

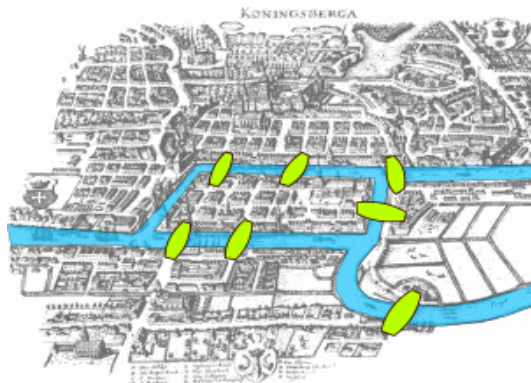


Figure 1: Map of Königsberg in Euler's time

The puzzle reached Swiss mathematician Leonhard Euler who found it trivial, but also intriguing. In 1741, he published a solution to the problem. Since he could not find a sufficient explanation using branches of math such as Algebra or Geometry, Euler solved it using an abstract process that led to a new branch of mathematics called “Graph Theory,” where he applied a new mathematic model called a graph.

1.2 Euler's Approach and Solution

According to Euler, physical maps did not matter. Rather, the important element to consider was which land masses were connected by bridges. He transformed the problem into a network of vertices and connecting edges, known today as a **connected graph**, with each vertex representing a land mass and each edge a connecting bridge.

For the purpose of our explanation, we establish some definitions:

- **Graph:** A set of vertices which are connected by edges.
- **Vertex:** A node on a graph.
- **Edge:** A link connecting a pair of vertices.
- **Degree of a vertex:** The number of edges incident to a vertex.
- **Walk:** A finite alternating list of vertices and edges.
- **Trail:** A walk with no repeated edges.
- **Path:** A walk with no repeated vertices.
- **Circuit:** A closed walk (starts and ends at the same vertex) that contains at least one edge and does not contain a repeated edge.

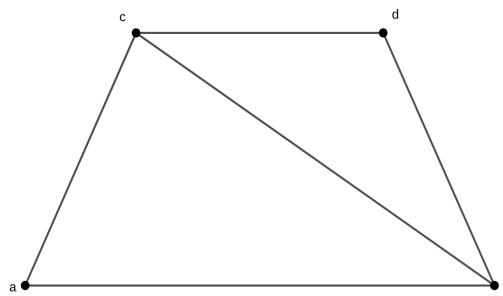


Figure 2: A graph with 4 vertices and 5 edges

For example, in the figure above, we have:

$$\begin{aligned}\deg(a) &= 2 \\ \deg(b) &= 3 \\ \deg(c) &= 3 \\ \deg(d) &= 2\end{aligned}$$

Now, returning to Euler's discourse...

Euler denoted each district with capital letters: A, B, C, and D. He also observed that Königsberg had one district that was connected by five bridges, while the other three regions were connected by three bridges (as shown in the figure below).

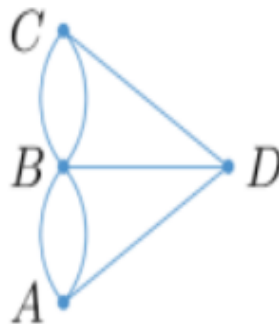


Figure 3: Euler's graph theory representation of the map of Königsberg

The graph above could be used to translate the Königsberg problem into the following question:

Is it possible to start at A, B, C or D, traverse each edge exactly once, and end at the starting point?

Finding such a possibility would be the equivalent of finding the solution to the puzzle.

We can see that when we start from the vertex A and go to B, C, and D, we always have to use one edge to get in and another edge to get out from each vertex. That signifies that we must have an even number of edges leading into and out of the vertices B, C and D. However, in our scenario, $\deg(C) = \deg(D) = 3$, and $\deg(B) = 5$. Hence, given the odd degrees of B, C and D, there is no such route that starts and ends at A and leads across all bridges only once. For example, as one of my possible trip routes, I can begin my travel at A and walk to B, then C and next I can go back to B and proceed back to A (my journey's starting point). However, since I need to visit D as well, I am forced to use the third "outbound" edge of A. Now, I cannot return to A because there is no "inbound" edge I could possibly use that leads me to A and has not been traveled already. Similarly, considering vertices other than A as a starting and ending point, we can arrive at the same conclusion. (Readers can verify this by attempting to trace the graph without lifting a pen and redrawing on line.)

Thus, it is impossible to find a closed path that satisfies the required condition of the Königsberg puzzle.

In honor of Euler, such a closed path, with no isolated vertices and where each edge was passed exactly once, became known as an **Euler circuit**.

1.3 Euler's Theorem

Generating one rule to solve puzzles with arbitrary number of bridges and land masses led to the following theorem:

1.3.1 Statement

Let the graph $G(V, E)$ be an undirected graph (in that an edge can be traversed in both directions) with no isolated vertices.

G has an Euler circuit if and only if:

1. G is connected.
2. Every vertex has an even degree.

1.3.2 Proof

First, we show that if G has an Euler circuit, then for all $v \in V$, the trail $T \{v_0, v_1, \dots, v_{k-1}, v_k\}$ with $v_k = v_0$, G is connected.

Trail T visits each vertex of G at least once (it is possible to revisit a vertex), and since every vertex appears somewhere along the route as a part of sequence of adjacent vertices and edges of

G , we can deduce that G is connected. Since the route passes through any given vertex v of G as the process of entering and departing by using distinct edges that lead to and from the vertex, we can see that the degree of each vertex v is even.

Conversely, let's consider a connected graph G with every vertex having an even degree (assuming the vertex set of G is nonempty). To show that G contains an Euler circuit, we proceed by induction on the number of edges n in G .

In the base case, there are two edges, and since the degree is even, then these edges must be between the same vertices. Therefore, an Euler circuit is immediately represented.

Now, we show the statement holds for fewer than n edges.

Let's consider a connected graph G with vertices of even degrees, and v_0 as our starting vertex of G . Now, let's pick any sequence of adjacent edges and vertices, without repeated edges containing v_0 , beginning and ending at v . Let's call the resulting circuit C .

If we see that C includes every edge of G , then we are done.

If this is not the case, we proceed to traverse C , and we keep removing its edges until we get to the vertex where there are no more outgoing edges, and this vertex must be v_0 . (We are stuck here because by assumption there are no vertices with an odd degree that would allow us to depart.)

In the process of removing edges, we end up with a subgraph of G , which we can denote by G' . As we traversed C , we have removed an even number of edges from each vertex, so G' has vertices of an even degree. Though it may not be connected, each component subgraph of G' is connected, and each vertex of these components has an even degree.

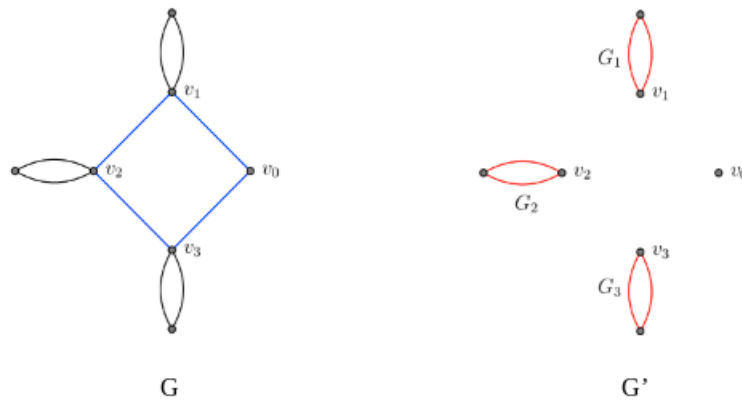


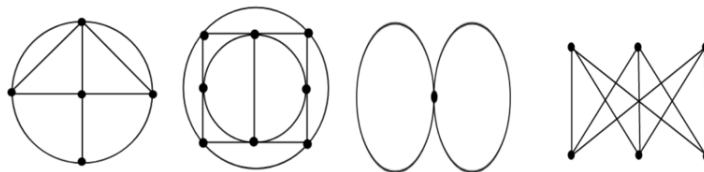
Figure 4: The graph G before and after removing edges

In addition, each of these Euler circuits has at least one vertex in common with G' . Next, we may choose any sequence of adjacent vertices and edges of G' . For example, starting at v_0 and then arriving at v_1 , moving along the Euler circuit of G_1 , and then moving to v_2 and completing the Euler circuit of G_2 , and then to v_3 and so on, until we get to the starting vertex because the degree of v_0 is an even integer and hence, there must be another edge that brings us back to v_0 . That way, we can reconstruct an Euler circuit for G , noting that our algorithm must terminate because G has finite number of edges.

1.4 Exercises in Finding a Euler's Circuit

[Solutions to all exercises included at the end.]

1. Which graph from the following collection represents an Eulerian circuit?



2. An architect wants to add additional bridges to the blueprint of Königsberg city. How many should she draw to make an Eulerian circuit?
3. Leonhard wants to visit the Western states of the United States. He wants to start and finish in the same state crossing each border exactly once. Is this possible?



1.5 Concluding Thoughts

Euler's intellect and vision to begin with a specific and perhaps banal problem but derive from it a new theory through a generalization process is indeed remarkable and inspiring. Today, graph theory has become an important branch of mathematics because web pages connected by links, social media pages, family charts, and the lattices formed by atoms are all modeled by graphs.

2 The Mathematics Behind the Game of Hex

“Problems worthy

of attack

prove their worth

by hitting back.” (Piet Hein)

2.1 Introduction

Piet Hein, a Danish mathematician and writer, and John Nash, an American mathematician and economist, invented the game of Hex independently in 1942 and 1948 respectively. Initially, in Copenhagen's Niel Bohr's Institute of Theoretical Physics, the game was called Polygon because of the diamond-shaped board used by players. In Princeton University, where Nash was an undergraduate student, the game was known as "John" (because the hexagonal pattern of tiles found in the bathroom created a suitable area to play). Ultimately, however, in 1952, the game was renamed Hex when a Salem game board company Parker Brothers, Inc. (mostly recognized for Monopoly) started its circulation. In 1953, Claude Shannon and E. F. Moore created a first Hex-playing analog machine at the Bell Telephone Laboratories. The general public was introduced to the game by Martin Gardner in his "Mathematical Games" column in *Scientific American* in 1959.

2.2 Let's Play!

The premise of the game is relatively simple. The board is composed of an 11×11 hexagonal grid, and two players alternate turns by claiming hexagons on the board. One player "owns" two opposite sides of the board, and the other "owns" the remaining two sides. (In our picture below, the colors are blue and red.) The objective of the game is to make a continuous path of "owned" hexagons from one side to the other.

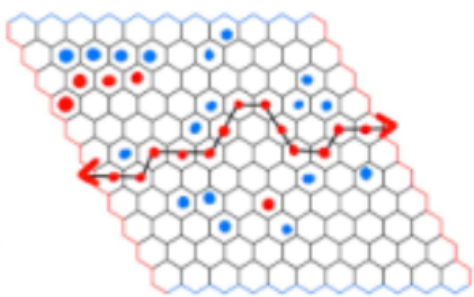


Figure 5: A sample game of Hex

2.3 The Hex Theorem

One particularly interesting component of Hex is that the game never ends in a draw; a player wins if and only if the other loses.

This Hex Theorem can be proved by interpreting the board as a planar graph, with vertices as corners of every hexagonal cell as shown in below figure.

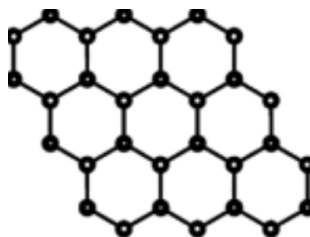


Figure 6: The Hex board as a planar graph

To prove the theorem, we begin with a graph lemma:

2.3.1 Lemma

Any finite graph with vertices of degree at most 2 is a union of disjoint components, and each component is either:

- an isolated vertex
- a simple cycle (defined as a closed path where middle vertices are not visited twice)
- a simple path (no vertices are repeated).

2.3.2 Proof

We proceed by induction.

Given a graph G with N vertices, where each vertex has degree of at most 2 and the maximum number of edges N , we can denote a graph with k edges as G_k .

For G_0 , our base case, all the nodes are isolated, as we have N vertices and 0 edges.

When a graph has $n + 1$ edges, we randomly choose an edge (s, v) to remove. The vertices s and v now have a degree at most 1, since they had a degree at most 2 initially.

Therefore, s and v cannot be on any cycles. By assumption, g_n is the union of disjoint isolated nodes, simple cycles, and simple paths.

When we add (s, v) back into the graph, the components that were disjoint from s and v in g_n remain the same by the addition of (s, v) , and the vertices s and v are now either on the same simple path or cycle.

Therefore, G_{n+1} is also the union of disjoint isolated vertices, simple cycles, and simple paths.

Thus, the lemma is true for all g_k with $0 \leq k \leq N$.

2.3.3 Proof of the Hex Theorem

Now, let's substitute the two colors in the game with Xs and Os, and add four vertices u_1 , u_2 , u_3 and u_4 at the four corners of the main graph and connected to the graph by four edges e_1 , e_2 , e_3 , and e_4 .

Every tile will be marked with either an X or O. Also, opposite sides will be changed to O-O' and X-X' respectively, as shown below.

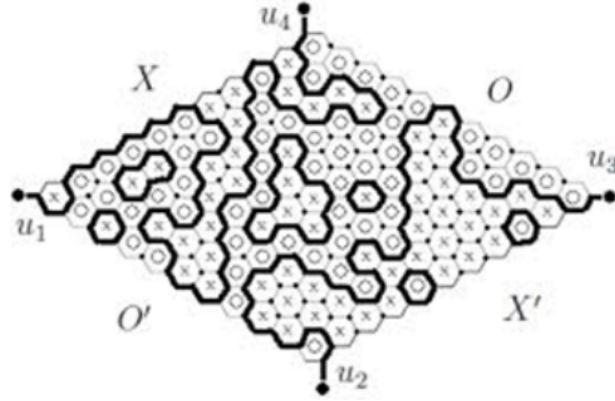


Figure 7: Adjusted Hex board with Xs and Os

Following our changes in representation, we can restate the Hex Theorem in the following way:

If every tile of the Hex board is marked either X or O, then there is either an X-path connecting regions X and X' or an O-path connecting regions O and O'.

In order to prove the Hex Theorem, it suffices to show that there is a simple path connecting any two corner vertices.

Step 1: Let's construct a subgraph $G'(V, E')$ containing the same vertices and the subset of the edges that lie between X and O tile faces.

Step 2: The corner vertices u_1, u_2, u_3 , and u_4 have degree 1.

Step 3: If a vertex is surrounded by two same patterned tiles and one differently patterned tile, then it has two edges. Thus, each vertex has degree of 0 or 2.

Step 4: By proof of the lemma above, if a graph has vertices with degree at most 2, the graph is a union of disjoint components: isolated vertices, simple cycles, or simple paths.

Step 5: Thus, there exist two paths connecting two of the corner endpoint vertices. Hence, there is a winning path for one of the players.

2.4 Zermelo and the Theory of Games

Since we have proven that Hex cannot end in a draw, one of the players must have a winning strategy. In other words, independently of how the opponent plays, with a finite number of moves, and the absence of chance, the player's strategy will result in a win. This forms the basis of Zermelo's Theory, which is often cited as the first mathematical analysis of graph theory.

In his paper "On an Application of Set Theory to the Theory of the Game of Chess" (1913), German mathematician Ernst Zermelo analyzed chess as an example of such finite, two-player, zero-sum games (where one player's gain corresponds with another player's loss) to figure out the optimal strategies for both players that decide both players' payoff. It is important to note that modern game theory was marked by John Von Neumann and Oskar Morgenstern's proof and

characterization of mixed equilibria in zero-sum games. John Nash subsequently generalized the analysis beyond zero-sum games to equilibrium in games that presented more complex situations, where people interact strategically and purposefully to maximize their advantage.

To show why Zermelo's theory is true, let's assume that the first player does not have a winning strategy and initiates a move. In this case, the second player prevents the first from winning by moving strategically in response. This preemptive set of moves guarantees that the second player cannot lose because she is presented with at least a drawing strategy. Now, since the game cannot end in draw, this drawing strategy is equivalent to forcing a win. Hence, she has a winning strategy.

In the game of Hex, the two players take turns, make a strategic finite number of moves without the element of chance, and cannot tie. Thus, Zermelo's theory posits that one of the two players must have a winning strategy.

2.5 The Strategy-Stealing Argument

Additionally, John Nash determined another interesting component of the game: the first player has always a winning strategy. (My younger sister mercilessly exploited this assertion when playing against me.) In 1949, Nash proved the strategy-stealing argument using the *reductio ad absurdum*, which can succinctly be explained as follows:

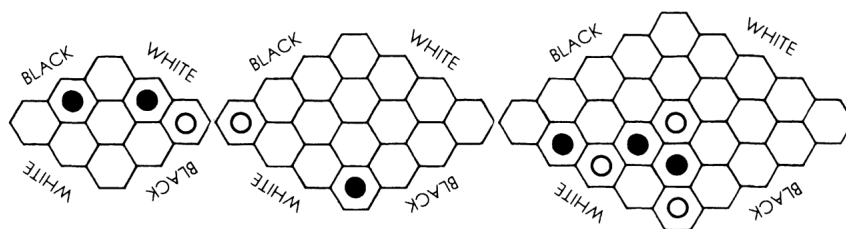
1. Assume that the second player has a winning strategy.
2. The first player makes a move by marking a cell anywhere on the board.
3. After the second player's move, the first player purposefully ignores his already claimed cell and pretends that the opponent's move is the initial one. This extra claimed cell on the board does not interfere with the player's game and serves only as an advantage.
4. In effect, the first player "steals" the second player's winning strategy, as he now masquerades as the second player.
5. This is a contradiction to our starting assumption that the second player has a winning strategy.
6. Thus, discarding the assumption, we can conclude that there is a winning strategy for the first player (remembering that a draw is impossible).

2.6 State of Play

Since the first player has a strong advantage in Hex, players are allowed to implement a swap rule, which allows for the second player to take the first player's initial move and claim his cell. This rule is often utilized during Hex tournament at Computer Olympiad organized by the International Committee of Mathematical Games (ICMG) since 2000.

2.7 Conclusion

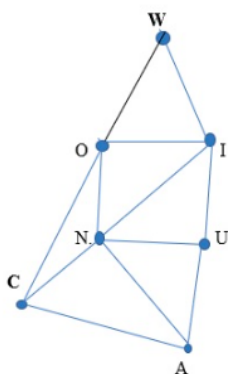
There are numerous online games available for the readers who would like to simply try Hex. For those readers who would like a challenge, Martin Gardner offered three Hex problems devised by Hein with the objective of finding first move that would ensure the win for "white."



Hex is a beautiful game with a simple set of rules yet full of critical tactical and strategic intricacies. With a variety of openings and 2.4×10^{56} legal positions (exceeding that of chess!), as Piet Hein fittingly put it in his grook, Hex indeed can become a problem that is worthy of attack.

Solutions to Exercises in Finding a Euler's Circuit

1. The third graph represents an Euler circuit because it is the only graph that is connected with vertices with an even degree.
2. The architect can draw two bridges in order to make an Euler circuit (for example, in Figure 1: another bridge that connects landmasses C and D, and one that connects landmasses A and B)
3. It is impossible.



The graph contains 4 vertices with an odd degree.

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